

Form, Height & *Glass.*

Which architectural parameters drive energy performance — and by how much? 768 simulated buildings, 8 parameters, one answer.

768

simulated buildings

12

building forms

8

architectural parameters

RESEARCH QUESTION

Which architectural design parameters matter most for energy performance? Using UCI Energy Efficiency Dataset and Random Forest modelling, this study ranks all 8 variables by predictive importance.

BEST VS WORST SCENARIO · HEATING LOAD

✓ BEST



B7 · Cubic
3.5m · 0% glass

11.9

kWh/m² HL

✗ WORST



B5 · Long bar
7m · 40% glass

49.7

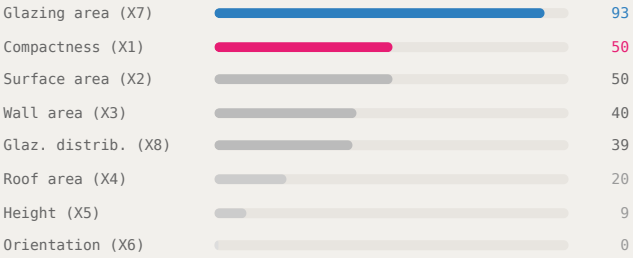
kWh/m² HL

4.2×

difference in heating load between the most and least efficient scenario — same dataset, same climate.

KEY FINDINGS

What drives energy load?



+78%

Heating load increase: 0% → 40% glazing (14.3 → 25.4 kWh/m²). Glass is the architect's most powerful energy lever.

2.4×

7m buildings use 2.4× more heating energy than 3.5m equivalents — same form, same glazing.

0 / 100

Orientation RF score — N/E/S/W difference stays below 0.12 kWh/m². Safely ignored at early design stage.

PARAMETERS & SIMULATION

8 inputs, 2 outputs

INPUT VARIABLES — X1 TO X8			
ID	Parameter	Range	RF
X7	Glazing area	0–40%	93
X1	Compactness	0.62–0.98	50
X2	Surface area	514–808 m²	50
X3	Wall area	245–416 m²	40
X8	Glazing distrib.	0–5 categ.	39
X4	Roof area	110–220 m²	20
X5	Height	3.5m / 7m	9
X6	Orientation	N/E/S/W	0

Y1 – Heating Load

6–50 kWh/m²

Y2 – Cooling Load

11–48 kWh/m²

ECOTECT SIMULATION PIPELINE

- ① Geometry

12 building forms as orthogonal volumes. Constant floor area 771.75 m² at 2 heights (3.5m & 7m) → 24 base geometries.
- ② Glazing

4 glazing ratios (0%, 10%, 25%, 40%) × 5 distribution patterns (uniform + N/E/S/W) applied to each geometry.
- ③ Thermal simulation

Annual simulation per config. Setpoints: 19°C heating / 26°C cooling. Athens climate. Admittance method.
- ④ Output

Results normalised to kWh/m². 4 orientations applied → 768 rows total.

B7 Cubic
3.5m · 0%



11.9

kWh/m² HL

B5 Long bar
7m · 40%

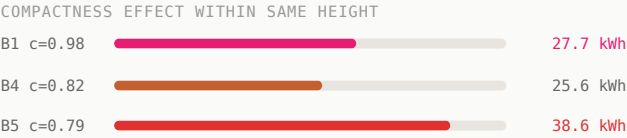
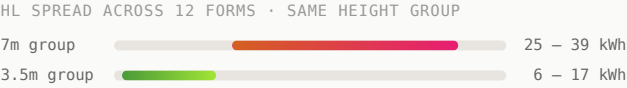
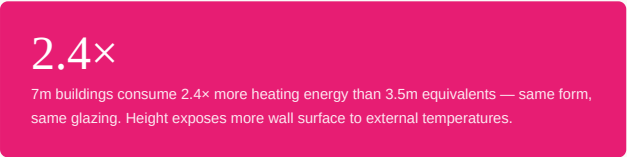
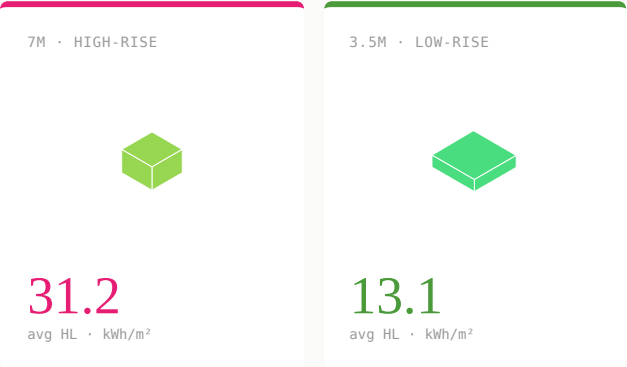


49.7

kWh/m² HL

HEIGHT VS FORM

Height dominates;
form is secondary



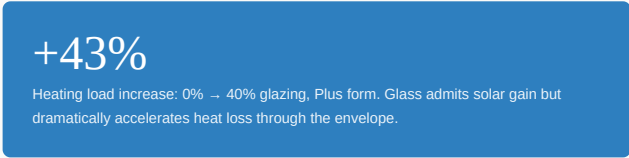
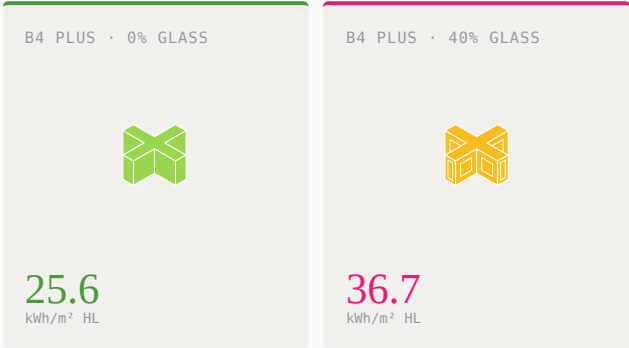
Counter-intuitive: taller buildings are less efficient even when compact. A 3.5m L-form outperforms a 7m cube in heating load — height beats form at every direct comparison.

ALL 12 FORMS · 4 GLAZING RATIOS · HEATING LOAD (KWH/M²)

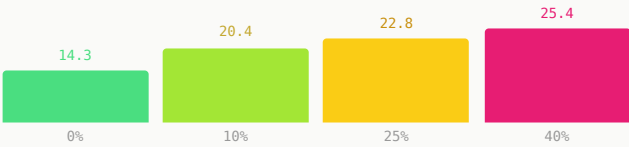


THE GLAZING EFFECT

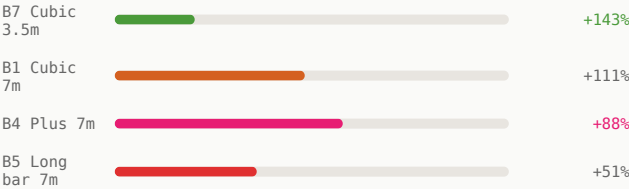
Same form,
radically different glass



MEAN HL · ALL 12 FORMS · BY GLAZING RATIO



GLAZING EFFECT BY BUILDING · 0% VS 40% HL



Glass is the single most powerful thermal lever available to the architect. Controlling glazing area at early design stage yields more savings than any choice of form or orientation.